



Assignment 1.2: System Security Requirements Report

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Executive Summary

Design World is a multinational engineering and design firm that depends heavily on the protection of its design data. As the company expanded, its earlier decentralized technology model created significant security risk. Employees were using different systems, tools, and connectivity methods, which made it difficult to consistently protect data, control access, and maintain security oversight. After Design World experienced targeted attacks and the theft of original design work, the organization moved to a private cloud model where employees access cloud-hosted virtual machines through VPN and perform work inside a protected environment.

This System Security Requirements Report defines the security requirements for Design World's sandbox mirror environment. The sandbox is designed to represent one production region and includes a segmented cloud architecture with a public subnet, business subnet, website/DMZ subnet, and transit VLAN. This design is intended to reduce direct exposure of internal systems while supporting secure remote access for Design World's organization.

The primary security concern for Design World is protecting its intellectual property. The report identifies the systems that store, process, and support access to this data as critical to the business. Because Design World operates with a fully remote workforce, the organization must rely on a secure and well-managed cloud environment to prevent unauthorized access, reduce the risk of data theft, and maintain business continuity.

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The security requirements in this report define how that environment should be secured and managed over time. Rather than focusing on individual controls, the requirements work together to establish a stronger security posture across the full environment. They help ensure that access is controlled, sensitive data is protected, system activity can be reviewed, and the environment can be tested and improved as Design World continues to mature its security program.

Overall, this report provides Design World with a clear security baseline for reviewing, testing, hardening, and documenting the sandbox environment. It connects the company's business risks to specific security objectives and technical requirements, while also preparing the environment for the next project phases, including vulnerability assessment, penetration testing, security hardening, post-hardening validation, and development of the Information System Security Plan. The end goal is to ensure that Design World's cloud environment can support secure remote operations and protect intellectual property,

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1. Engagement Overview

1.1 Purpose and Scope

This report establishes the system security requirements for Design World's sandbox mirror environment and defines the security baseline that will guide the remaining phases of the engagement. Its purpose is to translate Design World's business risks, stakeholder protection needs, and operating constraints into a set of technical security requirements that can be validated, implemented, and maintained across the environment. In keeping with systems security engineering principles, the report connects stakeholder concerns such as protection of intellectual property, secure remote access, and service availability to specific security objectives and system level requirements.

The scope of this report is limited to the Linode hosted sandbox that mirrors a single Design World production region, including the business subnet, the website subnet, the public access layer, and the transit and services VLAN. Systems addressed within scope include the domain controller, file server, Ubuntu workstation, CentOS web server, OpenVPN server, identity and access mechanisms, routing paths, and supporting administrative access flows. This report does not assess live production systems, employee owned endpoint devices outside the VPN connection point, or future phases of validation and hardening beyond defining the baseline requirements that those later activities will measure against.

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1.2 Engagement Context

Design World operates as a multinational engineering firm specializing in artistic structural support systems. The organization emerged from a merger of three independent firms and historically operated with decentralized IT-employees selected their own hardware, software, and connectivity. This approach became untenable as the company scaled beyond 100 employees and became a target for sophisticated industrial espionage.

Following multiple successful attacks that resulted in theft of proprietary designs and operational disruption, Design World partnered with Four Point Cyber Consulting to deploy a private enterprise cloud solution. Under the current model:

- All employees work remotely via Virtual Private Network (VPN)
- All work is performed within cloud-hosted Virtual Machines (VMs)
- Sensitive data remains within the protected perimeter
- Access is centrally managed through Active Directory

While the solution is operational, it has not been formally evaluated, documented, or validated against enterprise security requirements, creating the need for this engagement.

1.3 Regulatory and Compliance Driver

Design World's customer base includes U.S. government entities, which necessitates alignment with:

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- NIST SP 800-171 (Protecting Controlled Unclassified Information in Non-federal Systems)
- ISO 27000 series (Information Security Management Systems)
- Federal contracting security requirements

The deliverables from this engagement will support Design World's ability to demonstrate security due diligence, maintain competitive positioning for government contracts, and protect intellectual property against state-sponsored and criminal threat actors.

1.4 Roles and Responsibilities

Team Member	Role	Responsibilities
Christian Calipusan	Project Lead	Led the development of security requirements, reviewed technical controls, and aligned the document with project objectives.
Shivam Patel	Security Analyst	Reviewed system components, business needs, and operational context.
Joshua Remington	Security Engineer	Confirm the requirements are technically realistic for the Windows, Linux, VPN, and cloud environment
Mikailou Nikiema	Network Engineer	Define segmentation, VPN, routing, and access-path requirements accurately.

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2. Infrastructure Architecture

2.1 Network Topology

The Design World sandbox environment mirrors a single production region deployed on Linode infrastructure. The architecture applies defense in depth through network segmentation. The Public subnet contains the OpenVPN server, the Business subnet contains identity and data systems, and the Website/DMZ subnet contains the public-facing web server. The Transit VLAN supports VPN routing, DNS forwarding, and internet gateway functions.

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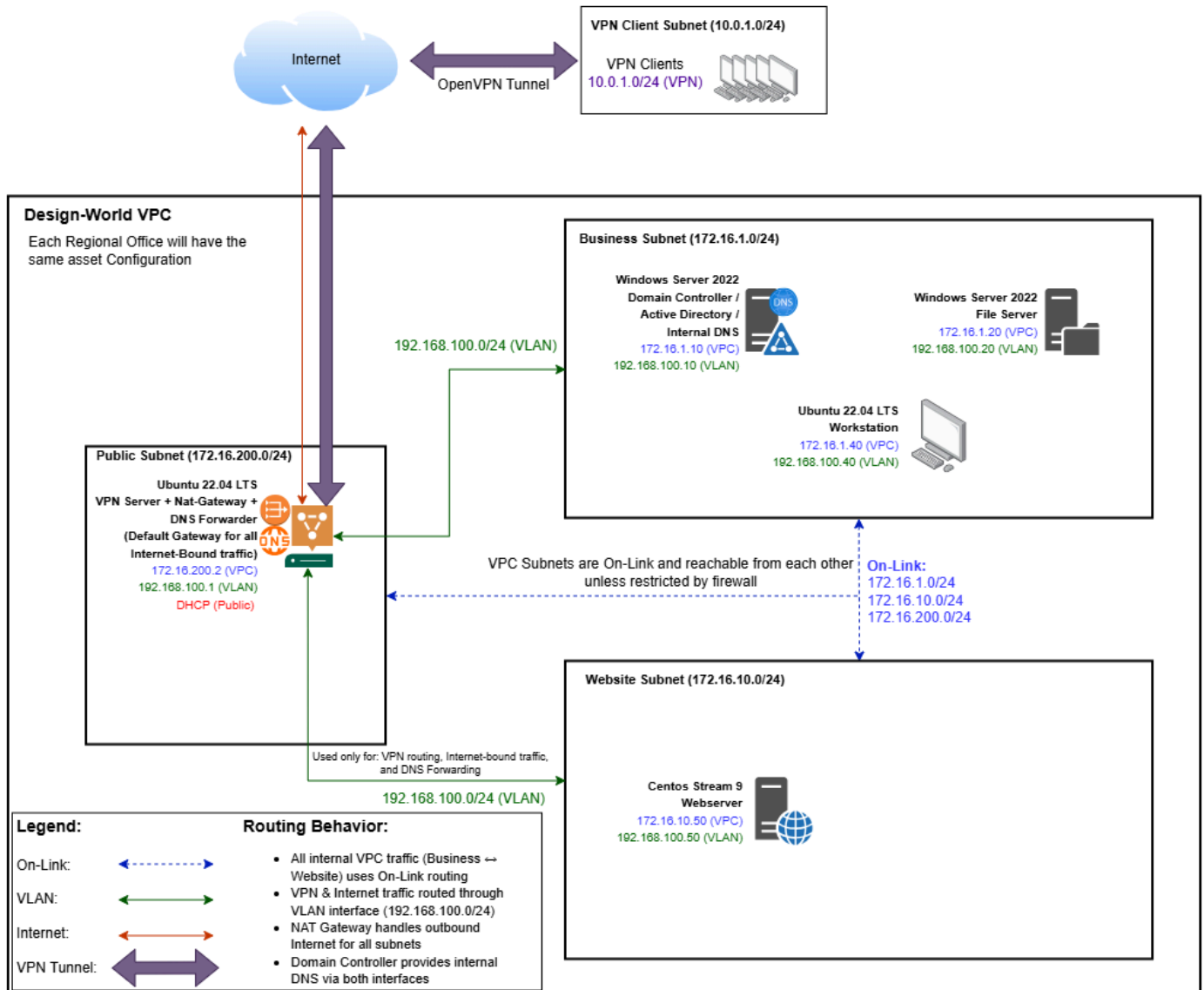


Figure 1: Network Topology Diagram

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2.2 System Inventory

Asset Type	Operating System	IP Address	Security Role
Domain Controller	Windows Server 2022	172.16.1.10	Identity and access management
File Server	Windows Server 2022	172.16.1.20	Sensitive Data Repository
Ubuntu Workstation	Ubuntu 22.04	172.16.1.40	Management and internal operations
Web Server	CentOS Stream 9	172.16..10.50	Public-facing customer interface
OpenVPN Server	Linux-based	172.16.200.2	Administrative access gateway

2.3 Access Control Architecture

All administrative access follows a mandatory path:

1. An external administrator establishes an encrypted VPN tunnel to the OpenVPN server at 172.16.200.2.
2. Traffic routes through the Transit VLAN at 192.168.100.x.
3. The Transit VLAN pivots traffic to the internal Business or Website subnets.
4. Direct public internet access to internal assets is prohibited by design.

User authentication is centralized through Active Directory. All remote employees must use VPN access before interacting with Design World systems or data.

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3. Stakeholder Analysis

3.1 Stakeholder Identification

Stakeholder	Primary Concern
DW Executive Leadership	Business continuity, intellectual property protection, competitive advantage
Remote Employees	Secure access to tools and data, reliable access to VMs and file services
System Administrators	System administrators Maintainable systems, secure administrative workflows, incident response capability
Government Customers	Regulatory compliance, supply chain assurance, data protection evidence
FourPoint Cyber Consulting	Deliverable quality, technical accuracy, successful engagement execution
Legal and Compliance	Legal and compliance Regulatory positioning, liability management, contractual evidence
Finance	Return on security investment and breach cost avoidance
Human Resources	Onboarding, offboarding, and access lifecycle support

3.2 Critical Assets

Asset Class	Sensitivity	Primary Threat	Required Protection
Intellectual property and design files	Critical	Espionage and theft	Confidentiality
Active Directory database	Critical	Compromise or manipulation	Integrity and availability
Domain Controller	Critical	Outage or compromise	Availability and integrity
File Server data repositories	High	Unauthorized access	Confidentiality and integrity
VPN gateway	Critical	Outage or bypass	Availability and integrity

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Administrative credentials	Critical	Theft or misuse	Confidentiality and integrity
System baseline configurations	High	Unauthorized modification	Integrity
DNS and routing infrastructure	High	Manipulation or outage	Integrity and availability
Web Server	Medium	Defacement or exploitation	Integrity and availability

3.4 Stakeholder Protection Needs

- Business leadership needs prevention of further intellectual property theft, operational continuity for the remote workforce, security due diligence for government contracts, and reduced reputational and financial impact from incidents.
- Employees need reliable access to the VPN, virtual machines, and file services, as well as clear policies and procedures that replace informal guidance.
- Government customers need evidence of NIST SP 800-171 and ISO 27000 alignment, supply chain security assurance, and incident response capability.
- Administrators need secure and auditable administrative access paths, configuration control, change management, and detection and response capabilities.

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4. Scope and Assumptions

4.1 Scope

This project focuses on a security review, evaluation, testing, and documentation of Design World's sandbox mirror environment. This sandbox mirror environment is a replica of a single production region. Four Point Cyber Consulting is responsible for performing the following activities within the Linode-hosted VPC environment:

- Conducting a formal Security Engineering Review of the current architecture and configuration
- Performing a Vulnerability Assessment and Penetration Test (pre- and post-hardening)
- Developing and submitting Project Change Requests (PCRs) and a Security Hardening Plan
- Implementing system hardening across the network environment
- Developing Design World's Information System Security Plan (ISSP) with supporting policies aligned with NIST SP 800-171 and ISO 27000
- Delivering final project documentation and a formal presentation

The sandbox environment includes segmented subnets connected via a transit and services VLAN. Access will be provided via OpenVPN from a Kali Linux VM. Access will be limited to the sandbox mirror environment and does not include live production

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systems or client endpoint devices beyond the VPN connection point. The Service Level Agreement (SLA) defines the responsibilities for Design World and Four Point Cyber Consulting.

4.2 Assumptions

The following core assumptions were provided via the Statement of Work (SOW):

- All Design World Systems are in a private cloud operated by a small US company in three worldwide regions
- The hardware is virtual Intel-based systems utilizing a combination of Windows and Linux operating systems
- All access is controlled by VPN access, Cloud Access, and all user access is through Active Directory
- All communication between sites is proprietary.
- All employees work remotely
- All employees were given VPN access that they are required to use
- The company is broken into four functional areas: Design, Engineering, Marketing, and Business Management
- All system design, operating instructions, and guidelines were given out over Zoom meetings with the designers

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5. Threats, Risks, and Security Concerns

5.1: Threats

Design World has been targeted by industrial espionage due to an increase in international recognition. Attacks facing Design World included physical break-ins and targeted hacking of remote designers' systems. Financial loss due to stolen designs led to the creation and deployment of the private enterprise cloud solution. Threat actors are assumed to be nation-state-sponsored groups, corporate competitors, and organized cyber criminals motivated by DW's intellectual property.

5.2: Strategic and Operational Risks

- **Intellectual Property Theft:** Exfiltration or loss of sensitive data remains the primary business risk with potential for competitive disadvantages and client loss
- **Multi-Regional Exposure:** Operations spanning the US, EU, and PRC introduce complexity in regulations, data storage, and threats
- **Remote Workforce:** The 100% remote access increases the attack surface and introduces more risk for VPN endpoints and client devices.

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- **Remote Training:** Reliance on Zoom-based guidance increases the likelihood of misconfiguration, human error, and misunderstandings of security practices.
- **Insufficient Documentation:** Lack of proper security documentation and resources decreases the prevention of social engineering and human-error-caused incidents.
- **Compliance and Contractual Risk:** Failure to demonstrate adequate controls may jeopardize government and certain civilian contracts.

5.3 Technical and Architectural Risks

- **Single Point of Failure:** Compromise of the OpenVPN server could allow lateral movement into the internal subnets through the Transit VLAN
- **Network Segmentation Gaps:** Inadequate firewall rules or VLAN controls may allow lateral movement between the subnets
- **Identity and Access Management:** Active Directory can be targeted to gain access to privileged credentials or control over domain access
- **Data Storage Exposure:** Weak access control or encryption may allow easy access to sensitive files at a central location
- **Hybrid Environment:** Multiple operating systems increase the demand for patching, hardening, and monitoring

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- **Internet-Bound Services:** Reliance on the Transit VLAN for updates and DNS forwarding may allow for command-and-control or inbound attack vectors

6. Security Objectives

- **SO-01: Control access into the environment**

The environment should only be accessible through approved and trusted entry points. Direct public internet access to internal business and website subnets should be blocked, and users or administrators should connect through an approved VPN before connecting to internal systems.

- **SO-02: Reduce unnecessary communication between network segments**

Network traffic between subnets should be limited to only what is needed for the environment to function. By controlling traffic through the Transit VLAN and allowing only required services such as Active Directory synchronization, SMB file sharing, and DNS forwarding, the environment reduces the chance of unauthorized movement between systems.

- **SO-03: Secure administrative access paths**

Administrative access should be protected through encrypted and controlled connections. Any administrative activity that requires pivoting into the internal

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environment should use the approved tunnel and Transit VLAN path instead of exposing internal systems directly.

- **SO-04: Use centralized identity for access control**

Access to virtual Intel-based systems should be managed through Active Directory whenever possible. This helps ensure that authentication is consistent, user activity can be tied back to a central identity source, and local accounts are reserved only for emergency recovery situations.

- **SO-05: Protect sensitive design data at rest**

Sensitive intellectual property, including original artistic structural designs stored on the file server, should be protected from unauthorized access. Volume-level encryption helps ensure that the data remains protected even if the underlying storage is exposed or compromised.

- **SO-06: Keep sensitive work inside the protected cloud environment**

The cloud environment should be configured so that users can perform their work without downloading sensitive files locally. This helps reduce the risk of data leakage and keeps project materials inside the controlled cloud perimeter.

- **SO-07: Harden systems based on their role and exposure**

Each system should be securely configured based on what it does and how exposed it is. Domain controllers, public-facing assets, and VPN gateways each have different risks, so the hardening approach should reflect the operating system, system role, and level of exposure.

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- **SO-08: Maintain visibility through vulnerability testing**

The environment should support regular vulnerability scanning and security validation. Systems need to be reachable from the Kali VM so that weaknesses can be reviewed, evaluated, and tested as part of the project's security assessment process.

- **SO-09: Protect sensitive Design World data**

Design World's intellectual property and design files should be protected from unauthorized access, exposure, or loss while stored and used within the cloud environment.

- **SO-10: Monitor access and system activity**

The environment should provide visibility into user access, administrative actions, and system activity so security events can be reviewed and investigated when needed.

- **SO-11: Maintain availability of critical services**

Core services such as VPN access, Active Directory, DNS, file storage, and the web server should remain available to support Design World's remote workforce and business operations.

- **SO-12: Support recovery from approved system baselines**

Systems should be recoverable using approved and controlled configurations, such as golden images, to ensure restored systems remain consistent with the secure architecture.

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- **SO-13: Validate the security of the environment**

The environment should be tested through vulnerability assessments, penetration testing, and post-hardening reviews to confirm that security controls are working as intended.

- **SO-14: Preserve approved network and access boundaries**

The environment should maintain the approved segmentation model and prevent users or systems from bypassing VPN, Transit VLAN, or subnet security boundaries.

- **SO-15: Document security changes and decisions**

Security decisions, hardening actions, testing results, and architecture changes should be documented so the environment can be maintained and improved over time.

7. Security Requirements

7.1 Access Control

- **SR-01 - Role-Based Access Control (RBAC):** Permissions must be mapped to the four functional areas (Design, Engineering, Marketing, Business Management) and the user's position if necessary (NIST SP800-171 - 3.1.1).

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- **SR-02 - Least Privilege:** Limit employee access within the "Business" and "Website" subnets to only what is necessary for their specific roles (NIST SP800-171 - 3.1.5)
- **SR-03 - Identity Consolidation:** All systems (including the CentOS Web Server and Ubuntu Workstations) must use **LDAP/Active Directory** for identification and authentication NIST SP800-171 - 3.5.1, 3.5.2, 3.5.4).
- **SR-04 - Multi-Factor Authentication (MFA):** Mandatory MFA for the OpenVPN entry point to mitigate the risk of stolen credentials (NIST SP800-171 - 3.5.3).
- **SR-05 - Privileged Access Management (PAM):** Administrative actions on the Domain Controller and Transit VLAN must require a separate, high-privilege account (NIST SP800-171 - 3.1.6).

7.2 Network Security

- **SR-06 - Remote Access Security:** All remote access must be established via an encrypted **OpenVPN tunnel**. DW must monitor and control these sessions at the regional private cloud entry points (NIST SP800-171 - 3.1.12).
- **SR-07 - VPN Security:** DW must **disable split tunneling** on user devices to prevent remote systems from simultaneously connecting to the corporate cloud and unsecured external networks (NIST SP800-171 – 3.13.7).

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- **SR-08 - DMZ Security:** The DMZ (Website Subnet) must be isolated; it should not be able to initiate connections to the Business Subnet (NIST SP800-171 – 3.13.5, 3.13.6).
- **SR-09 - Micro-segmentation:** Implement firewall rules within the Transit VLAN to restrict traffic between the File Server and the Web Server to specific, required ports only (NIST SP800-171 – 3.13.5, 3.13.6)

7.3 Endpoint and Server Security

- **SR-10 - System Hardening:** All Virtual Machines (VMs) must be deployed from hardened templates (STIG or CIS benchmarks) to ensure consistency across the mirror environment (NIST SP 800-171 – 3.4.1).
- **SR-11 - Host-Based Firewalls:** Active host-based firewalls must be configured on all endpoints to prevent lateral movement if a single VM is compromised (NIST SP800-171 – 3.4.7).
- **SR-12 - Endpoint Detection & Response (EDR):** DW must install EDR on all endpoints to monitor, detect, alert, and respond to malicious activities in real-time (NIST SP800-171 – 3.6.2, 3.14.2, 3.14.3, 3.14.7).

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7.4 Data Protection

- **SR-18 - Encryption at Rest:** All design files on the **DW-FS** (File Server) must be encrypted to protect against data theft from the physical cloud host (NIST Sp800-171 – 3.1.19).
- **SR-19 - Data Loss Prevention:** The cloud environment must block unauthorized data transfer functions between the VM and the employee’s local hardware (NIST SP800-171 – 3.13.4).
- **SR-20 - Secure Transit:** FIPS validated encryption must be used for all data moving across the Transit VLAN and VPN (NIST SP800-171 – 3.13.8).

7.5 Monitoring and Audit

- **SR-21 - Centralized Logging:** All logs from the OpenVPN server and Domain Controller must be forwarded to a central repository for review (NIST SP800-171 – 3.3.1).
- **SR-22 - Log Protection:** Protect audit information from unauthorized access or modification to ensure it remains reliable for future incident responses NIST SP800-171 – 3.3.5)
- **SR-23 - Audit Trails:** Log all user activities and modifications to sensitive design files on the File Server (NIST SP-800-171, 3.3.2).

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- **SR-24 - Alerting:** DW must monitor and control all remote sessions to detect anomalies such as failed login attempts and “impossible travel” (NIST SP800-171 – 3.1.12).

7.6 Availability and Recovery

- **SR-25 - Backup Policy:** DW must conduct regular backups (hourly incremental, daily differential, weekly full backups) of the Domain Controller and File Server to allow for rapid recovery (NIST SP800-171 – 3.8.9).

7.7 Assurance and Validation

- **SR-26 - Vulnerability Scanning:** DW must conduct regular vulnerability scans (for example biweekly) on all systems in the cloud environment (NIST SP800-171 – 3.11.2).
- **SR-28 - Penetration Testing:** DW must conduct quarterly penetration tests to ensure the implemented security controls are effective (NIST SP800-171 – 3.12.1).
- **SR-29 - Incident response and Reporting (3.6.2):** DW must develop a plan to track, document, report, and respond to security incidents (NIST SP800-171 – 3.6.2)

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8. Traceability Matrix

Security Objectives (SO)	Description	Related Security Requirements (SR)	NIST Controls SP800-171
SO-01	Access Control of the environment	SR-04, SR-06	3.5.3, 3.1.12
SO-02	Limit inter-subnet communications	SR-08, SR-09	3.13.5, 3.13.6
SO-03	Secure administrative access paths	SR-05, SR-06	3.1.6, 3.1.12
SO-04	Use centralized identity	SR-03	3.5.1, 3.5.2, 3.5.4
SO-05	Protect sensitive design data at rest	SR-18	3.1.19
SO-06	Keep sensitive work inside the cloud	SR-07, SR-19	3.13.7, 3.13.4
SO-07	Harden systems based on role	SR-10, SR-11	3.4.1, 3.4.7
SO-08	Maintain visibility via testing	SR-26	3.11.2
SO-09	Protect Design World data	SR-18, SR-19, SR-20	3.1.19, 3.13.4, 3.13.8
SO-10	Monitor access and system activity	SR-21, SR-23, SR-24	3.3.1, 3.3.2, 3.1.12
SO-11	Maintain availability of services	SR-25	3.8.9
SO-12	Support recovery from baselines	SR-10, SR-25	3.4.1, 3.8.9
SO-13	Validate environment security	SR-26, SR-28	3.11.2, 3.12.1

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SO-14	Preserve network boundaries	SR-08, SR-09, SR-11	3.13.5, 3.13.6, 3.4.7
SO-15	Document security changes	SR-29	3.6.2

9. Timeline for Deliverables

11 May 2026 – Security Requirements Report (Scoping Contract)
 18 May 2026 – Vulnerability Assessment Plan
 23 May 2026 – Vulnerability Assessment Report / Penetration Testing Plan
 01 Jun 2026 – Penetration Testing Report
 08 Jun 2026 – Security Hardening Plan (and PCRs)
 22 Jun 2026 – Security Hardening Report (after two weeks of hardening effort)
 29 Jun 2026 – Post Vulnerability Scan Report
 06 July 2026 – Post Penetration Testing Report
 20 July 2026 – ISSP Plan
 27 July 2026 – Capstone Project Report
 03 Aug 2026 – Capstone Project Presentation

10. Conclusion

Design World's operating model makes the security a business requirement rather than a technical preference because the company depends on remote access, centralized cloud services, and the protection of highly sensitive design data. The requirements defined in this report establish a clear foundation for controlling access, limiting lateral movement, protecting sensitive information, improving system visibility, and preserving the

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availability of critical services across the sandbox environment. They also provide a traceable link between Design World's security risks and the technical expectations that should govern the environment going forward.

As the first formal security engineering deliverable in the engagement, this report creates the baseline for subsequent assessment, testing, hardening, and the documentation activities identified in the Statement of Work. By defining security requirements before remediation and validation begin, Design World is better positioned to evaluate whether its cloud environment supports secure remote operations and provides protection proportionate to the value of its intellectual property and contractual obligations. A well defined requirements baseline also supports consistency across future security decisions and gives the organization a stronger foundation for long term governance, assurance, and compliance efforts.

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Role	Printed Name	Signature	Date
Project Lead	Christian Calipusan	<i>Christian Calipusan</i>	05/11/26
Security Analyst	Shivam Patel	<i>Shivam Patel</i>	05/11/26
Security Engineer	Joshua Remington	<i>Joshua Remington</i>	05/11/26
Network Engineer	Mikailou Nikiema	<i>Mikailou Nikiema</i>	05/11/26
DW CEO			